

$$1. P(X=k) = \frac{C_{39}^{5-k} C_{13}^k}{C_{52}^5} \quad (k=0,1,2,3,4,5)$$

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$$2. P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad P(X=1) = \lambda e^{-\lambda}, P(X=2) = \frac{\lambda^2}{2} e^{-\lambda}$$

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$$\Rightarrow \lambda = 2 \quad (\text{注意泊松分布要求}\lambda\text{为正数}) \quad P(X=4) = \frac{2^4}{4!} e^{-2}$$

$$3. p_d(d) = \begin{cases} \frac{1}{b-a}, & \text{当 } a \leq d \leq b \text{ 时} \\ 0, & \text{其它} \end{cases} \quad \text{圆面积的取值范围为 } [\frac{\pi a^2}{4}, \frac{\pi b^2}{4}]$$

$$\frac{\pi a^2}{4} \leq y \leq \frac{\pi b^2}{4} \text{ 时 } P(S \leq y) = P(\pi(\frac{d}{2})^2 \leq y) = P(d \leq 2\sqrt{\frac{y}{\pi}}) = \frac{1}{b-a} (2\sqrt{\frac{y}{\pi}} - a)$$

$$\text{求导得 } p_s(y) = \frac{1}{(b-a)\sqrt{\pi y}} \quad \text{因此 } p_s(y) = \begin{cases} \frac{1}{(b-a)\sqrt{\pi y}}, & \text{当 } \frac{\pi a^2}{4} \leq y \leq \frac{\pi b^2}{4} \text{ 时} \\ 0, & \text{其它} \end{cases}$$

$$E(S) = \int_{\frac{\pi a^2}{4}}^{\frac{\pi b^2}{4}} \frac{1}{(b-a)\sqrt{\pi y}} \cdot y dy = \frac{1}{(b-a)\sqrt{\pi}} \cdot \frac{2}{3} y^{\frac{3}{2}} \Big|_{y=\frac{\pi a^2}{4}}^{\frac{\pi b^2}{4}} = \frac{\pi(b^3 - a^3)}{12(b-a)}$$

$$E(S^2) = \int_{\frac{\pi a^2}{4}}^{\frac{\pi b^2}{4}} \frac{1}{(b-a)\sqrt{\pi y}} \cdot y^2 dy = \frac{1}{(b-a)\sqrt{\pi}} \cdot \frac{2}{5} y^{\frac{5}{2}} \Big|_{y=\frac{\pi a^2}{4}}^{\frac{\pi b^2}{4}} = \frac{\pi^2(b^5 - a^5)}{80(b-a)}$$

$$D(S) = E(S^2) - (E(S))^2 = \frac{\pi^2}{720} (b-a)^2 (4b^2 + 7ab + 4a^2)$$

$$4. (1) 1 = \int_{-\infty}^{+\infty} p(x) dx = \int_0^{+\infty} c e^{-x} dx = c \Rightarrow c = 1$$

$$(2) 1 = \int_{-\infty}^{+\infty} p(x) dx = \int_0^{+\infty} c x^\alpha e^{-\beta x} dx \xrightarrow{y=\beta x} \int_0^{+\infty} c \left(\frac{y}{\beta}\right)^\alpha e^{-y} \cdot \frac{1}{\beta} dy$$

$$= \frac{c}{\beta^{\alpha+1}} \int_0^{+\infty} y^\alpha e^{-y} dy = \frac{c \Gamma(\alpha+1)}{\beta^{\alpha+1}} \Rightarrow c = \frac{\beta^{\alpha+1}}{\Gamma(\alpha+1)}$$

$$(3) 1 = \int_{-\infty}^{+\infty} p(x) dx = \int_{-\infty}^{+\infty} \frac{c}{1+x^2} dx = \pi c \Rightarrow c = \frac{1}{\pi}$$

$$5. F(a-0) = \lim_{x \rightarrow a^-} F(x) = \lim_{n \rightarrow +\infty} P(X \leq a - \frac{1}{n}) = P(X < a) = P(X \leq a) - P(X = a)$$

$$P(a < X < b) = P(X < b) - P(X \leq a) = F(b-0) - F(a) \quad \text{因为 } \{X < a\} = \bigcup_{n=1}^{\infty} \{X \leq a - \frac{1}{n}\}$$

$$6. E(Y) = \int_{-\infty}^{+\infty} p(x) e^x dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \cdot e^x dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu-\sigma^2)^2}{2\sigma^2} + \frac{2\mu\sigma^2 + \sigma^4}{2\sigma^2}} dx$$

$$\xrightarrow{y=x-\sigma^2} e^{\frac{2\mu\sigma^2 + \sigma^4}{2\sigma^2}} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy = e^{\frac{2\mu\sigma^2 + \sigma^4}{2\sigma^2}} = e^{\frac{2\mu + \sigma^2}{2}}$$

$$E(Y^2) = \int_{-\infty}^{+\infty} p(x) e^{2x} dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \cdot e^{2x} dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu-2\sigma^2)^2}{2\sigma^2} + 2\sigma^2 + 2\mu} dx$$

$$\xrightarrow{y=x-2\sigma^2} e^{2\sigma^2 + 2\mu} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy = e^{2\sigma^2 + 2\mu}$$

$$D(Y) = E(Y^2) - (E(Y))^2 = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1)$$

$$7. 1 = \int_{-\infty}^{+\infty} p(x) dx = \int_0^{+\infty} A x e^{-\frac{x^2}{2\sigma^2}} dx = \left(-\sigma^2 A e^{-\frac{x^2}{2\sigma^2}} \right) \Big|_0^{+\infty} = \sigma^2 A \Rightarrow A = \frac{1}{\sigma^2}$$

$$E(X) = \int_{-\infty}^{+\infty} x p(x) dx = \int_0^{+\infty} \frac{x^2}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}} dx = \int_0^{+\infty} (-x) d(e^{-\frac{x^2}{2\sigma^2}}) \\ = \left(-x e^{-\frac{x^2}{2\sigma^2}} \right) \Big|_{x=0}^{+\infty} + \int_0^{+\infty} e^{-\frac{x^2}{2\sigma^2}} dx \stackrel{x=\sqrt{2}\sigma y}{=} \sqrt{2}\sigma \int_0^{+\infty} e^{-y^2} dy = \sigma \sqrt{\frac{\pi}{2}}$$

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 p(x) dx = \int_0^{+\infty} \frac{x^3}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}} dx = \int_0^{+\infty} (-x^2) d(e^{-\frac{x^2}{2\sigma^2}}) \\ = \left(-x^2 e^{-\frac{x^2}{2\sigma^2}} \right) \Big|_{x=0}^{+\infty} + 2 \int_0^{+\infty} x e^{-\frac{x^2}{2\sigma^2}} dx = \left(-2\sigma^2 e^{-\frac{x^2}{2\sigma^2}} \right) \Big|_{x=0}^{+\infty} = 2\sigma^2$$

$$\text{var}(X) = E(X^2) - (E(X))^2 = \left(2 - \frac{\pi}{2} \right) \sigma^2$$

$$P(X > E(X)) = P(X > \sigma \sqrt{\frac{\pi}{2}}) = \int_{\sigma \sqrt{\frac{\pi}{2}}}^{+\infty} \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}} dx = \left(-e^{-\frac{x^2}{2\sigma^2}} \right) \Big|_{x=\sigma \sqrt{\frac{\pi}{2}}}^{+\infty} = e^{-\frac{\pi}{4}}$$

8. 如果 $Y=F(X)$ 服从 $[0,1]$ 上的均匀分布, 则 $P(a \leq Y \leq b) = P(a \leq F(x) \leq b) = b-a$

由 $F(x)$ 不连续知它存在间断点 x_0 . 则 $F(x)$ 在 $x=x_0$ 左侧不连续 (注意, 分布函数有个性质, 就是它在每一点处都必定右连续)

设 $y_1 = \lim_{\delta \rightarrow 0^+} F(x_0 - \delta)$, $y_2 = F(x_0)$. 考虑 $P(0 \leq F(x) \leq \frac{2y_1 + y_2}{3})$ 和 $P(0 \leq F(x) \leq \frac{y_1 + 2y_2}{3})$

由 y_1, y_2 的定义知二者应相等. 这与 Y 服从均匀分布矛盾.

$$9. Y \text{ 取值为 } (0, +\infty). y > 0 \text{ 时 } P(Y < y) = P(\frac{1}{2} m X^2 < y) = P(0 < X < \sqrt{\frac{2y}{m}}) = \int_0^{\sqrt{\frac{2y}{m}}} \frac{4x^2}{\alpha^3 \sqrt{\pi}} e^{-\frac{x^2}{\alpha^2}} dx$$

$$\text{求导得 } p_Y(y) = \sqrt{\frac{2}{m}} \cdot \frac{1}{2\sqrt{y}} \cdot \frac{4}{\alpha^3 \sqrt{\pi}} \cdot \frac{2y}{m} e^{-\frac{2y}{m\alpha^2}} = \frac{4\sqrt{2y}}{\alpha^3 m \sqrt{m\pi}} e^{-\frac{2y}{m\alpha^2}} (y > 0). p_Y(y) = 0 (y \leq 0)$$

10. 该点的圆心角 θ 服从 $[0, 2\pi)$ 上的均匀分布. 落点的横坐标为 $X = R \cos \theta$

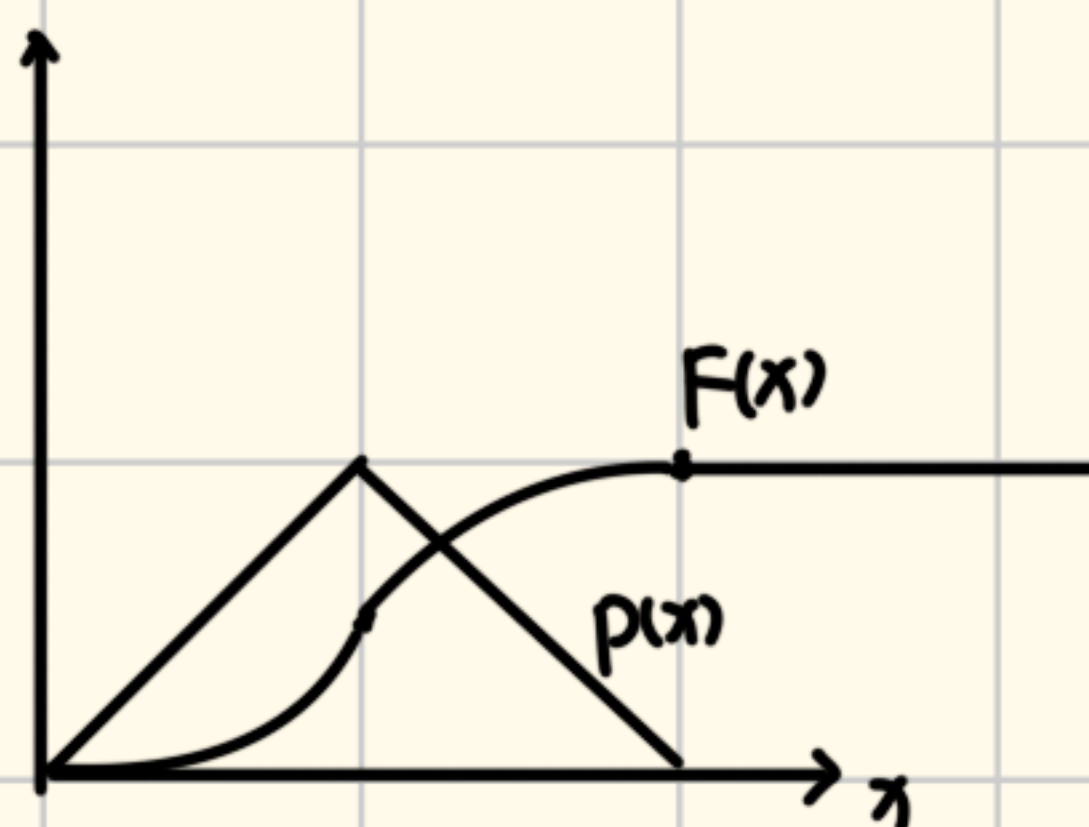
$$-R \leq x \leq R \text{ 时 } P(X \leq x) = P(\cos \theta \leq \frac{x}{R}) = P(\arccos \frac{x}{R} \leq \theta \leq 2\pi - \arccos \frac{x}{R}) = \frac{\pi - \arccos \frac{x}{R}}{\pi}$$

$$\text{求导得 } p_X(x) = \frac{1}{\pi \sqrt{R^2 - x^2}} \quad (-\frac{x}{R} \leq x \leq \frac{x}{R}). \quad p_X(x) = 0 \text{ (其它)}$$

$$E(X) = \int_{-\infty}^{+\infty} x p(x) dx = \int_{-R}^R \frac{x}{\pi \sqrt{R^2 - x^2}} dx = 0$$

$$11. x < 0 \text{ 时 } F(x) = 0. 0 \leq x < 1 \text{ 时 } F(x) = \int_0^x x dx = \frac{x^2}{2}$$

$$1 \leq x \leq 2 \text{ 时 } F(x) = \frac{1}{2} + \int_1^x (2-x) dx = -\frac{1}{2} x^2 + 2x - 1. x > 2 \text{ 时 } F(x) = 1$$



12. 设 $40 = k\sigma$, 要求 $P(\mu - k\sigma \leq X \leq \mu + k\sigma) \geq 0.80$

$$\text{而 } P(\mu - k\sigma \leq X \leq \mu + k\sigma) = P\left(-k \leq \frac{X - \mu}{\sigma} \leq k\right) = 2\Phi(k) - 1$$

标准正态分布的分布函数

$$\Rightarrow \Phi(k) \geq 0.90. \text{ 查表得 } k_{\min} \approx 1.28. \text{ 故 } \sigma_{\max} \approx 31.2$$

$$13. E(X) = \int_{-\infty}^{+\infty} x p(x) dx = \int_{-\infty}^{+\infty} x \cdot \frac{1}{2} e^{-|x|} dx = 0 \text{ (奇函数)}$$

$$14. E(X) = \int_{-\infty}^{+\infty} x p(x) dx = \int_{-\infty}^{+\infty} (c+y) p(c+y) dy \stackrel{x=c+z}{=} \int_{-\infty}^{+\infty} (c-z) p(c-z) (-dz) = \int_{-\infty}^{+\infty} (c-z) p(c-z) dz \\ = \frac{1}{2} \left(\int_{-\infty}^{+\infty} (c+x) p(c+x) dx + \int_{-\infty}^{+\infty} (c-x) p(c-x) dx \right) = \frac{1}{2} \cdot \int_{-\infty}^{+\infty} 2c p(c+x) dx = c$$

$$15. E(X) = \sum_{k=0}^l k \frac{C_D^k C_{N-D}^{n-k}}{C_N^n} = \sum_{k=1}^l \frac{D C_{D-1}^{k-1} C_{N-D}^{n-k}}{C_N^n} = \frac{D C_{N-1}^{n-1}}{C_N^n} = \frac{nD}{N} \left(\begin{array}{l} \text{第二个等号是 } k C_D^k = k \cdot \frac{D!}{k!(D-k)!} = D C_{D-1}^{k-1} \\ \text{第三个等号是 } \sum_{k=0}^m C_p^k C_q^{m-k} = C_{p+q}^m \end{array} \right)$$

$$E(X^2) = \sum_{k=0}^l k^2 \frac{C_D^k C_{N-D}^{n-k}}{C_N^n} = \sum_{k=0}^l (k^2 - k) \frac{C_D^k C_{N-D}^{n-k}}{C_N^n} + \sum_{k=0}^l k \frac{C_D^k C_{N-D}^{n-k}}{C_N^n}$$

$$\sum_{k=0}^l (k^2 - k) \frac{C_D^k C_{N-D}^{n-k}}{C_N^n} = \sum_{k=2}^l \frac{D(D-1) C_{D-2}^{k-2} C_{N-D}^{n-k}}{C_N^n} = \frac{D(D-1) C_{N-2}^{n-2}}{C_N^n} = \frac{D(D-1)n(n-1)}{N(N-1)}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{n(N-n)D(N-D)}{N^2(N-1)} = \underbrace{D \cdot \frac{n}{N} \left(1 - \frac{n}{N}\right) \cdot \frac{N-D}{N-1}}_{\text{模仿二项分布来解释: 次数} \times \text{概率} \times (1 - \text{概率}) \times \text{修正因子}}$$

$$16. x < 0 \text{ 时 } p(x) = \frac{1}{2} e^x, 0 \leq x < 1 \text{ 时 } p(x) = 0, x > 1 \text{ 时 } p(x) = \frac{1}{4} e^{-\frac{x-1}{2}}$$

$$E(X) = \int_{-\infty}^0 x \cdot \frac{1}{2} e^x dx + \int_1^{+\infty} x \cdot \frac{1}{4} e^{-\frac{x-1}{2}} dx = \int_{-\infty}^0 \frac{x}{2} d(e^x) + \int_1^{+\infty} \left(1 - \frac{x}{2}\right) d(e^{-\frac{x-1}{2}})$$

$$= \left(\frac{x}{2} e^x\right) \Big|_{x=-\infty}^0 - \int_{-\infty}^0 \frac{1}{2} e^x dx + \left(-\frac{x}{2} e^{-\frac{x-1}{2}}\right) \Big|_{x=1}^{+\infty} + \int_1^{+\infty} \frac{1}{2} e^{-\frac{x-1}{2}} dx = 1$$

$$17. (1) E(X) = 10 \cdot 0.4 + 11 \cdot 0.3 + 12 \cdot 0.2 + 13 \cdot 0.1 = 11$$

$$(2) E(Y) = 50 \cdot 3 \cdot 0.4 + 50 \cdot 2 \cdot 0.3 + 50 \cdot 1 \cdot 0.2 + 50 \cdot 0 \cdot 0.1 = 100$$

$$(3) E(\tilde{Y}) = 50 \cdot 3 \cdot 0.5 + 50 \cdot 2 \cdot 0.4 + 50 \cdot 1 \cdot 0.1 = 120 \quad \text{增加 } 20.$$

$$18. E(X) = \sum_{k=0}^{+\infty} k P(X=k) = \sum_{k=1}^{+\infty} \sum_{i=1}^k P(X=k) = \sum_{i=1}^{+\infty} \sum_{k=i}^{+\infty} P(X=k) = \sum_{i=1}^{+\infty} P(X \geq i)$$

$$19. \text{ 设函数 } I(x) = \begin{cases} 1, & \text{当 } x \geq \varepsilon \text{ 时} \\ 0, & \text{当 } x < \varepsilon \text{ 时} \end{cases} \text{ 则 } g(\varepsilon) I(x) \leq g(x). \text{ 取期望即得待证式}$$

$$20. Y \text{ 取值为 } (0, 1]. \text{ 当 } 0 < y \leq 1 \text{ 时, } P(Y \leq y) = P(\sin X \leq y) = P(0 < X \leq \arcsin y \text{ 或}$$

$$\pi - \arcsin y \leq X < \pi) = \int_0^{\arcsin y} \frac{2x}{\pi^2} dx + \int_{\pi - \arcsin y}^{\pi} \frac{2x}{\pi^2} dx = \frac{1}{\pi^2} \left[(\arcsin y)^2 + \pi^2 - (\pi - \arcsin y)^2 \right]$$

$$\text{求导得 } P_Y(y) = \frac{2}{\pi \sqrt{1-y^2}} \quad (0 < y \leq 1). \quad P_Y(y) = 0 \quad (y \leq 0 \text{ 或 } y > 1)$$

$$21. \quad y \leq 0 \text{ 时 } P(Y \leq y) = 0. \quad 0 < y \leq 1 \text{ 时 } P(Y \leq y) = \frac{y}{5}. \quad 1 < y < 3 \text{ 时 } P(Y \leq y) = P(1 < X < 3 \text{ 且 } X \leq y, \text{ 或 } X \leq 1) = \frac{y}{5}$$

$$3 \leq y < 5 \text{ 时 } P(Y \leq y) = \frac{3}{5}. \quad y \geq 5 \text{ 时 } P(Y \leq y) = 1.$$

$$22. \quad x \leq 0 \text{ 时 } p_x(x) = 0. \quad 0 < x \leq 3 \text{ 时 } p_x(x) = \frac{2}{9}x. \quad x > 3 \text{ 时 } p_x(x) = 0. \quad E(X) = \int_0^3 x \cdot \frac{2}{9}x dx = 2$$

$$p \text{ 为位数 } z_p \text{ 满足 } \int_{-\infty}^{z_p} p_x(x) dx = p. \text{ 解得 } z_p = 3\sqrt{p}$$

$$23. \quad \text{对 } y \in \mathbb{R}, \quad P(Y \leq y) = P(\ln X \leq y) = P(X \leq e^y) = 1 - e^{-(\frac{e^y}{\eta})^\alpha}$$

$$\text{求导得 } p_Y(y) = -e^{-(\frac{e^y}{\eta})^\alpha} \cdot (-\frac{e^y}{\eta^\alpha}) \cdot \alpha e^{\alpha y} = \frac{\alpha}{\eta^\alpha} e^{\alpha y} - \frac{e^{\alpha y}}{\eta^\alpha}$$

24. 有一个巧妙的想法, 想象这个游戏每一轮比两局, 那么有 $2pq$ 的概率无事发生, 有 $1-2pq$ 的概率游戏结束. 所以这是一个几何分布. 套用几何分布期望公式 $E(X) = \frac{1}{1-2pq}$

得原题的 $E = \frac{2}{1-2pq}$. 分子是2是因为我们假想的游戏中的一局实际是两局

$$25. \quad a > 0 \text{ 时}, \quad P(Y \leq y) = P(aX + b \leq y) = P(X \leq \frac{y-b}{a}) = \int_{-\infty}^{\frac{y-b}{a}} p_x(x) dx$$

$$\text{求导得 } p_Y(y) = \frac{1}{a} p_x(\frac{y-b}{a}). \quad a < 0 \text{ 时同理}$$

$$26. \quad \sum_{x=1}^6 P(X=x) = 1 \Rightarrow C = \frac{1}{21}. \quad E(X) = \sum_{x=1}^6 x P(X=x) = \frac{13}{3}. \quad \text{中位数为 } 5$$

$$27. \quad (1) \quad \mu = 10, \sigma^2 = 4. \quad P(6 < X < 9) = P(\mu - 2\sigma < X < \mu) - P(\mu - \frac{1}{2}\sigma < X < \mu)$$

$$= P(-2 < \frac{X-\mu}{\sigma} < 0) - P(-\frac{1}{2} < \frac{X-\mu}{\sigma} < 0) = \Phi(-\frac{1}{2}) - \Phi(-2) = 0.29$$

$$(2) \quad P(13 \leq X \leq 15) = P(\mu < X \leq \mu + \frac{5}{2}\sigma) - P(\mu < X < \mu + \frac{3}{2}\sigma) = \Phi(\frac{5}{2}) - \Phi(\frac{3}{2}) = 0.06$$

$$28. \quad P(X=k) = \frac{b^k}{k!} e^{-b}. \quad \text{要找最小的 } m, \text{ 使 } \sum_{k=0}^m P(X=k) \geq 0.99 \text{ 查表, } m=12$$

$$29. \quad 0 < p \leq \frac{1}{2} \text{ 时}, \quad p = \int_{-\infty}^{z_p} \frac{1}{2} e^x dx = \frac{1}{2} e^{z_p} \Rightarrow z_p = \ln(2p). \quad \frac{1}{2} < p < 1 \text{ 时 } z_p = -\ln(2(1-p))$$

$$30. \quad E(d) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{b}{\pi} \cdot \frac{v^2}{g} \sin 2\theta d\theta = \frac{bv^2}{\pi g} \cdot (\frac{1}{2} \cos 2\theta) \Big|_{\theta=\frac{\pi}{6}}^{\frac{\pi}{3}} = \frac{3v^2}{\pi g}$$

$$E(d^2) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{b}{\pi} \cdot \frac{v^4}{g^2} \sin^2 2\theta d\theta = \frac{3v^4}{\pi g^2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (1 - \cos 4\theta) d\theta = (\frac{1}{2} + \frac{3\sqrt{3}}{4\pi}) \frac{v^4}{g^2}$$

$$\text{var}(d) = E(d^2) - (E(d))^2 = (\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} - \frac{9}{\pi^2}) \frac{v^4}{g^2}$$

$$31. E(i) = \int_{900}^{1100} \frac{1}{200} \cdot \frac{120}{r} dr = \frac{3}{5} \ln \frac{11}{9}$$

$$E(i^2) = \int_{900}^{1100} \frac{1}{200} \cdot \frac{14400}{r^2} dr = \frac{4}{295} \quad \text{var}(i) = E(i^2) - (E(i))^2$$

$$32. E(X) = \int_0^{+\infty} \frac{1}{m!} \lambda^{m+1} e^{-\lambda} d\lambda = \frac{1}{m!} \Gamma(m+2) = m+1$$

$$E(X^2) = \int_0^{+\infty} \frac{1}{m!} \lambda^{m+2} e^{-\lambda} d\lambda = \frac{1}{m!} \Gamma(m+3) = (m+2)(m+1) \quad \text{var}(X) = m+1$$

由切比雪夫不等式 $(P(|X - E(X)| > \varepsilon) \leq \frac{1}{\varepsilon^2} \text{var}(X), \forall \varepsilon)$ 得 $P(|X - E(X)| > m+1) \leq \frac{1}{m+1}$

$$\Rightarrow P(0 < X < 2(m+1)) = P(|X - E(X)| < m+1) \geq \frac{m}{m+1}$$

$$33. \int_a^b a p(x) dx \leq E(X) = \int_a^b x p(x) dx \leq \int_a^b b p(x) dx$$

$$\begin{aligned} \text{var}(X) &= E\left(\left(X - \frac{a+b}{2} + \frac{a+b}{2} - E(X)\right)^2\right) = E\left(\left(X - \frac{a+b}{2}\right)^2\right) + 2\left(\frac{a+b}{2} - E(X)\right)E\left(X - \frac{a+b}{2}\right) + E\left(\left(\frac{a+b}{2} - E(X)\right)^2\right) \\ &= E\left(\left(X - \frac{a+b}{2}\right)^2\right) - \left(\frac{a+b}{2} - E(X)\right)^2 \leq E\left(\left(X - \frac{a+b}{2}\right)^2\right) \end{aligned}$$

$$\text{由 } |X - \frac{a+b}{2}| \leq \frac{b-a}{2} \text{ 得 } E\left(\left(X - \frac{a+b}{2}\right)^2\right) = \int_a^b \left(X - \frac{a+b}{2}\right)^2 p(x) dx \leq \int_a^b \left(\frac{b-a}{2}\right)^2 p(x) dx = \frac{(b-a)^2}{4}$$

模仿这道题: X, Y 是随机变量, 证明: $\min_{a, b} \{E(Y - (a + bx))^2\} = \text{var}(Y)(1 - \rho_{XY}^2)$

$$\begin{aligned} \text{证: } E((Y - (a + bx))^2) &= E\left([Y - E(Y)] - b[X - E(X)] - a + E(Y) - bE(X)]^2\right) \\ &= E((Y - E(Y))^2) + b^2 E((X - E(X))^2) + (a - E(Y) + bE(X))^2 - 2bE((Y - E(Y))(X - E(X))) \\ &= \text{var}(Y) + b^2 \text{var}(X) - 2b\rho_{XY}\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)} + (a - E(Y) + bE(X))^2 \\ &= (1 - \rho_{XY}^2)\text{var}(Y) + (\rho_{XY}\sqrt{\text{var}(Y)} - b\sqrt{\text{var}(X)})^2 + (a - E(Y) + bE(X))^2 \end{aligned}$$

经验就是, 一方面照着取等条件放. 另一方面, 多试着把 X 拆成

$(X - E(X)) + E(X)$. 由于 $E(X - E(X)) = 0$, 所以展开后能消掉一些项